

# Comprehensive Sage Test Document

Sage Test Suite

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## Comprehensive Sage Test Document

This document combines multiple Sage test documents to verify the functionality of the Sage filter.

### Table of Contents

- Basic Sage Test
- Mathematics and Plotting
- Advanced Examples

### Basic Sage Test

#### Simple Calculation

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

#### Display Result

The sum of a and b is:  $5 + 7 = 12$

#### Code Example

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

### Mathematics and Plotting

#### Basic Computation

```
x = 2
y = 3
z = x + y
```

#### Plot Example

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

#### Matrix Operations

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

### Advanced Examples

#### 3D Plotting

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

## Symbolic Mathematics

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

## Interactive Widgets

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

## Conclusion

This document serves as a comprehensive test of Sage functionality in Markdown documents. It includes: - Basic calculations and variable assignments - Mathematical expressions and plots - Matrix operations - 3D visualizations - Symbolic mathematics - Interactive elements (for testing in non-PDF contexts)

To compile this document to PDF with Sage code execution, use:

```
pandoc --filter sage_filter_package -o combined_sage_tests.pdf combined_sage_tests.md
```